MIL-R-26/4D
11 May 1973
SUPERSEDING
MIL-R-26/4C
9 February 1968

MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (POWER TYPE)

STYLES RW55, RW56, RW67, RW68, AND RW69

ALL STYLES, EXCEPT RW56, ARE INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION. FOR REPLACEMENT ITEMS SEE 6.3.

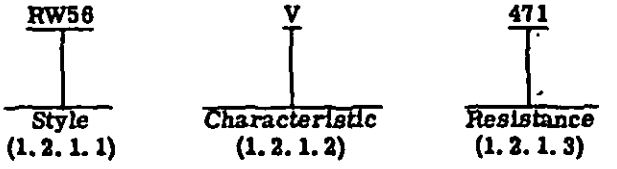
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for styles RW55, RW56, RW67, RW68, and RW69 resistors.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form:



1.2.1.1 Style. The style is identified by the two-letter symbol "RW" followed by a two-digit number; the letters identify power-type, wire-wound, fixed resistors, and the number identifies the size, wattage rating, and construction of the resistors.

1.2.1.2 Characteristic. The characteristic is identified by a single-letter that identifies the maximum continuous operating temperature (surface hot spot), the minimum insulation resistance value at the end of the moisture resistance test, and the resistance temperature characteristic in accordance with table I.

TABLE I. Characteristic.

Symbol	Maximum continuous operating temperature 1/	Minimum insulation resistance at end of moisture resistance test	Resistance temperature characteristic (percent per degree C)
	°C	Megohms	
V - - -	350	100	±.040 for resistance values below 20 ohms. ±.026 for resistance values of 20 ohms and above.

1/ This temperature is also the maximum permissible hot-spot surface temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:22Z
Check the source to verify that this is the current version before use.

1.2.1.3 Resistance. The nominal resistance expressed in ohms is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeroes to follow. Where fractional values of an ohm are required, the letter "R" shall be substituted for one of the significant digits to represent the decimal point. When the letter "R" is used, the last digit becomes significant, as shown in the following example:

R47 = .47 ohm
4R7 = 4.7 ohms.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-26 - Resistors, Fixed, Wire-wound (Power Type), General Specification for.

STANDARDS

MILITARY

MS90178 - Standard 24-Value Series Decade for Electronic Components of 5-, 10-, and 20-Percent Tolerances.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-26, and as specified herein.

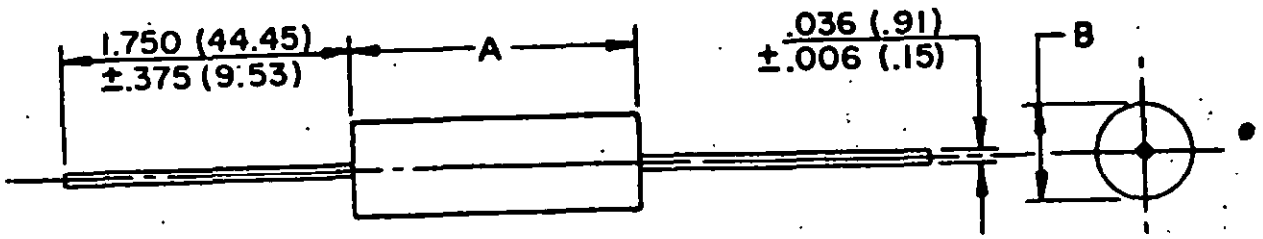
3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions as specified on figure 1, and as specified herein.

3.3 Power rating. The power rating shall be as specified in table II, based on full load operation at an ambient temperature of 25°C.

TABLE II. Power rating.

Resistor style	Power rating in watts	NATO style per NEPR No. 5	NATO power rating at 70°C (watts)
	Characteristic V		
RW55 - - - - - -	7	NRW08	6
RW56 - - - - - -	14	NRW09	12
RW67 - - - - - -	6.5	NRW16	5.6
RW68 - - - - - -	11	NRW17	9.5
RW69 - - - - - -	3	NRW18	2.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:22Z
Check the source to verify that this is the current version before use.



	RW55	RW56	RW67	RW68	RW69
A	1.375±.094 (34.93±2.39)	2.000±.094 (50.80±2.39)	1.000±.094 (25.40±2.39)	$1.875^{+.063}_{-.125}$ ($47.63^{+1.60}_{-3.18}$)	.500±.063 (12.70±1.60)
B	.469±.094 (11.91±2.39)	.469±.094 (11.91±2.39)	.313±.063 (7.95±1.60)	.344±.094 (8.74±2.39)	.188±.063 (4.78±1.60)

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are for general information only and are based upon 1 inch = 25.4 m m.
3. These styles shall be untapped.
4. Millimeters are in parentheses.
5. The end of the resistor body is the point where the diameter reduces to .06 inch.

FIGURE 1. Styles RW55, RW56, RW67, RW68, and RW69.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:22Z
Check the source to verify that this is the current version before use.

3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table III.

3.4.1 Standard resistance values. For standard resistance values, see MS90178.

TABLE III. Minimum and maximum nominal resistance value.

Resistor style	Resistance value		
	Minimum	Maximum 1/	Maximum 2/
	ohm	ohms	ohms
RW55 - - - - - -	0.10	5,100	---
RW56 - - - - - -	0.10	9,100	---
RW67 - - - - - -	0.10	3,600	8,200
RW68 - - - - - -	0.10	8,200	20,000
RW69 - - - - - -	0.10	910	2,000

1/ Based on use of 0.00175-inch nominal diameter wire (single layer wound).
2/ Based on use of 0.001-inch nominal diameter wire (single layer wound).

3.5 Resistance tolerance. Resistors of less than 1 ohm in resistance value shall have a tolerance of ±10 percent of the nominal resistance value. Resistors of 1 ohm and over shall have a tolerance of ±5 percent of the nominal resistance value.

3.6 Thermal shock. The resistor shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.7 Short-time overload. The resistor shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.8 High-temperature exposure. The resistors shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.9 Moisture resistance. The resistors shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.10 Low-temperature storage. The resistors shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.11 Life. The resistors shall not change in resistance in excess of ±(3 percent +.05 ohm).

3.12 Solderability. Solderability is applicable to this specification.

3.13 Terminal strength. The resistors shall not change in excess of ±(1 percent +.05 ohm).

3.14 Dielectric withstanding voltage. The resistors shall not change in resistance in excess of ±(.1 percent +.05 ohm).

3.15 Shock (medium impact). The resistors shall not change in resistance in excess of ±(.2 percent +.05 ohm).

3.16 Vibration (high frequency). The resistors shall not change in resistance in excess of ±(.2 percent +.05 ohm).

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:22Z
Check the source to verify that this is the current version before use.

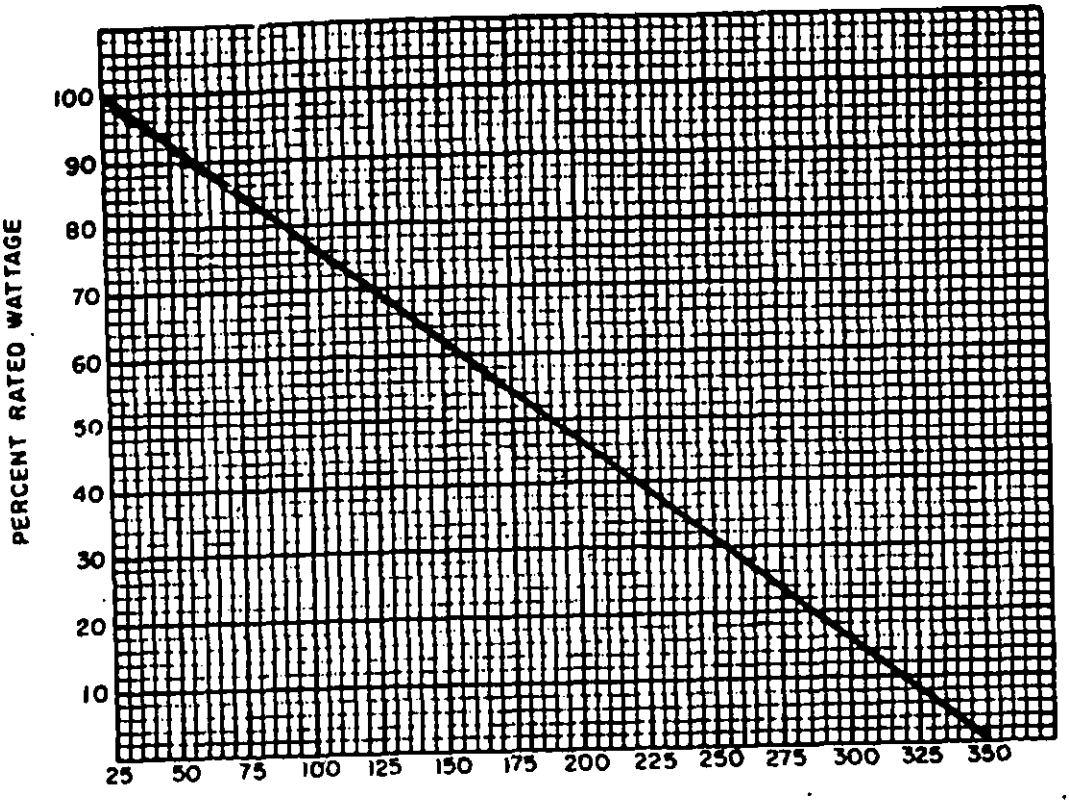


FIGURE 2. Derating curve for high ambient temperatures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:22Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-26, and as specified herein.

4.2 Barometric pressure (reduced). The magnitude of test voltage shall be 300 volts rms at 100,000 feet for styles RW55, RW56, RW67, and RW68. For style RW69, the test voltage shall be 200 volts rms at 100,000 feet.

4.3 Short-time overload. The short time overload shall be 5 times rated wattage for 5 seconds for style RW69, and 10 times rated wattage for 5 seconds for styles RW55, RW56, RW67, and RW68.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-26.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-26 are applicable to this specification.

6.2 Derating. Resistors shall be derated, when necessary, in accordance with figure 2.

* 6.3 Substitution data. Except for style RW56, the resistors specified herein are not for use in new design after the date of this specification. They are authorized for use in design contracts effective prior to or on the date of this specification, and to support existing military equipment. Resistors of MIL-R-39007 are preferred for new design. Although there are no MIL-R-39007 styles equivalent to the MIL-R-26/4 styles, the following substitutions can be made for the inactivated resistors:

MIL-R-26 styles	MIL-R-39007 detail specification	Substitute styles	Resistance tolerance
RW55	MIL-R-39007/10	RWR84	1% (F)
RW67	MIL-R-39007/10	RWR84	1% (F)
RW68	MIL-R-39007/7	RWR78	1% (F)
RW69	MIL-R-39007/11	RWR89	1% (F)

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI, MU, EL
Navy - EC, AS, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5905-0820-3)

☆U.S. GOVERNMENT PRINTING OFFICE: 1973-714-848/5488

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:22Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Communications - Electronics Command
ATTN: DRSEL-ED-TO
Fort Monmouth, NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:22Z
Check the source to verify that this is the current version before use.